

thylinao at BAREC Shared Task 2026: Ensembling at the Noise Floor

Maksim Silchenko

National University of Singapore
e1506804@u.nus.edu

Abstract

Team thylinao entered both tracks of the BAREC 2026 shared task on sentence-level Arabic readability with a deliberately plain system: a fourteen-member equal-weight ensemble of encoder fine-tunes, decoded to the 19-level scale by cutpoints interpolated halfway between validation-tuned thresholds and thresholds that match the blind score distribution to the training label prior. It scored 85.3 QWK on both tracks, second on both. This paper is about how those choices were measured. Three independent estimates place the noise floor of our offline evaluation between 0.09 and 0.44 QWK, as large as the margins that decided the final ranking. In three paired comparisons against the live board, only the smallest of the three offline estimates survived; the other two reversed in sign. We therefore proved that the Open and Strict tracks score identical gold, and made every decision worth a submission from a live board reading. Once the board showed the mean was capped, every further candidate came back a tie at the board’s resolution. The full ledger, including four refuted hypotheses and one withdrawn submission, is in the appendix.

1 Introduction

Midway through the BAREC 2026 campaign, our ensemble scored 85.44, 85.28 and 85.42 on the same test split while we varied nothing but the resolution of a threshold search grid. Chasing that anomaly produced the central finding of this paper: measured three independent ways, the noise floor of our offline evaluation sits between 0.09 and 0.44 QWK, while the margins that decided our two placings were 0.1 and 0.2 QWK. An evaluation cannot choose between changes smaller than its own noise, and ours did not. In three paired tests, two of three offline estimates reversed in sign on the blind leaderboard. We responded by rebuilding the campaign around the one instrument with

adequate resolution, first proving that the Open and Strict tracks score identical blind gold, then making every decision worth a submission from a live board reading. This paper describes the system that finished second on both tracks at 85.3 QWK, and the measurement discipline that chose it, which we believe is the part worth reusing.¹

The system is ordinary on purpose, and we report that plainness as a result: every fitted alternative we tried scored below plain equal weight (Section 5).

The first application of the rule came early. The public CAMEL-Lab readability checkpoints were fine-tuned on the BAREC corpus, which contains the shared task’s training split, so their training-split predictions are in-sample: solo QWK of 89 to 95 on train against 82 to 84 on clean holdouts. Any blend weight or cutpoint tuned on train would have been tuned on memorised labels, so we retired train as a tuning surface and adopted tune-on-validation, evaluate-once-on-test.

We contribute a noise-floor audit of offline evaluation on this task, with three independent estimators and three paired offline-versus-board tests on the true blind gold; a protocol for using a second track as a measurement instrument, including a cross-track parity proof and two competition mechanics that changed our decisions more than any model change did; a cutpoint calibration that was the one lever clearing the noise floor, worth 84.1 to 84.5 on the board; and an audited ledger of four refuted hypotheses and the improvements the ensemble absorbed without moving (Appendices A and B).

2 Background

BAREC grades Arabic sentences on a 19-level ordinal scale aligned to a pedagogical reading

¹Code, the fine-tuning grid and the per-run score arrays: <https://github.com/thylinao/barec-2026-readability>

progression (Elmadani et al., 2025b; Habash et al., 2025). The shared task evaluates systems with Quadratic Weighted Kappa (Elmadani et al., 2026). QWK penalises an error by the square of its distance in levels, so a system is judged on the shape of its prediction distribution at least as much as on per-sentence accuracy. That drove most of what we did, and it is visible in the final standings: two teams finished with far higher exact accuracy than ours and far lower QWK.

We entered the sentence-level Open and Strict tracks. Three results from the 2025 edition shaped our plan. Ensembling across loss families was the winning margin: the winner (Basem et al., 2025) combined four encoders under three loss families and reported 84.8 for its best single model rising to 87.5 for the ensemble, and the runner-up stacked seven regression base learners under a meta-classifier (Saeed et al., 2025). Ordinal framing and D3Tok morphological tokenisation are table stakes, worth roughly +3.1 and +1.0 on the organisers’ own baselines, and other entries explored ordinal heads, uncertainty-aware decoding and QWK surrogates (Nazzal, 2025; Abdou, 2025; Trigui, 2025; Mohamed Emad Elden, 2025). External data mostly did not help, and the organisers say so directly (Elmadani et al., 2025a): the winning system scored worse in Open than in Strict. That is why our Open and Strict systems are identical. We treated the Open track not as a licence to add outside corpora but as a second measurement of the same blind gold, which turned out to be worth more.

3 System Overview

3.1 Members and blend

The final system averages fourteen z-standardised member scores at equal weight, standardising each member’s validation and blind scores independently before averaging. Four seats are public BAREC fine-tunes released by others, nine are seed-averaged families of our own fine-tunes and one is a single run. Ours use the BAREC training split alone (Section 5.5). Members came from a cluster grid of 112 completed fine-tuning runs over 13 backbones, 6 loss families and three input views, combined with public checkpoints into a pool of 172 candidates (Appendix D).

3.2 The measurement chain

Before training anything we reimplemented the official scorer and gated the project on exact agreement, reproducing the official script string-for-string on the development fixture. A separate entry point pins the label set to [1, 19] for per-fold evaluation, where a sparse tail would otherwise silently build a smaller confusion matrix. The gate paid for itself in one upload: a submission whose local test QWK was 86.10 returned 86.10 on the development-phase board, which scores that released test split, all seven metrics matching, confirming that our grader is the board’s grader.

Preprocessing parity is the same kind of concern. The blind set is new sentences, so the corpus’s gold morphological views must be recomputed locally with CAMEL Tools (Obeid et al., 2020). Two failure modes here are silent: feeding raw text to a model that expects the Word view drops that member from 84 QWK to 50, and omitting only the character-normalisation step inside that pipeline drops exact match against gold from 99% to 31%. Neither raises an error; both produce a plausible submission.

The subtler version cost us a member. Our locally computed D3Tok matches the released gold at about 76%, because the released column used a licence-gated morphological database. Measured naively this appeared to drop the regression member from 84.68 to 82.75, so we shipped a word-only ensemble. That was wrong, and the error was in the standardisation, not the pipeline: the comparison took the member’s validation scores from the gold column while taking its blind scores from ours, and z-scoring cannot reconcile two input distributions. Recomputing validation through the same pipeline as blind made the member self-consistent, and adding it back was worth +0.3 on the board, more than every later calibration experiment combined. A member’s standardisation statistics and the scores they are applied to must come from one pipeline, end to end.

3.3 Half-match cutpoint calibration

Cutpoints tuned on validation put the blind predictions in the wrong place. The system predicted level 12 for about 8% of blind sentences where the corpus prior puts roughly 21%. The scores shift between validation and blind, so cutpoints fitted in validation score space land wrongly in blind score space, and quadratic weighting punishes the result-

ing systematic level shift.

The correction uses the only label-free target available, the training label histogram. Let r be the validation-tuned cutpoint vector, p the training label prior over levels 1 to 19, and s_b the blind blended scores. We ship

$$c = \text{sort}((1 - f)r + fQ(s_b, \text{cumsum}(p)_{1:18})) \quad (1)$$

with $f = 0.50$, where Q takes empirical quantiles of s_b at the cumulative prior boundaries. The shipped cutpoints sit halfway between what maximised QWK on validation and what would make the blind prediction histogram match the corpus prior.

We measured f on the live board, because blind has no labels and the test split does not exhibit the shift that motivates the correction. The curve is single-peaked: $f = 0$ gives 84.1, $f = 0.50$ gives 84.5, $f = 0.60$ gives 84.4, and $f = 1$ gives 84.0. Both endpoints are worse than the midpoint, by 0.4 and 0.5, because the two targets carry complementary and partly wrong information: the rounder knows the score-to-level mapping but assumes the wrong score distribution, while the prior match knows the right output distribution but discards the rounder’s ordering information. After five readings we stopped tuning f .

4 Experimental Setup

The release provides train (54,845 sentences over 1,518 documents), validation (7,310) and test (7,286) with gold labels, and a blind test of 8,077 sentences over 212 documents. Folds are built once with grouped stratified 5-fold cross-validation, grouped on document so topic and vocabulary do not leak, and serialised to a committed fold map that is never regenerated. Our go/no-go number is a nested out-of-fold QWK in which cutpoints are fitted on the other folds and applied to the held-out fold, so thresholds never see the rows they score. Every member shows validation below test by about 1.3 QWK, which correctly predicted the blind set would score nearer 85 than 86.

Calibrating the instrument. We estimated the noise of that protocol three ways. Holding data, members and weights fixed and varying only the threshold search grid between 150 and 500 candidates moves the same blend over a range of 0.345 QWK with standard deviation 0.094. Freezing the

change	offline	board
add MARBERTv2 (nested OOF)	+0.406	−0.1
document shrinkage $\alpha=0.1$	+0.22	−0.2
add D3Tok regression member	+0.086	+0.3

Table 1: Paired offline-versus-board comparisons. Each row is one change, estimated offline and then measured on the blind gold.

cutpoints and bootstrap-resampling the 7,286 test rows 400 times puts the sampling standard deviation of a single test measurement at about 0.44 QWK. Varying the fold-map seed, by contrast, returns bit-identical results, because the rounder snaps cutpoints to a fixed grid and small changes in the fitting rows often do not move the selected grid point. That third result is a trap worth naming: anyone estimating variance by reshuffling folds on a grid-search rounder will conclude, incorrectly, that their procedure is deterministic. Vary the grid instead. A paired bootstrap shrinks a naive +0.52 gain, from adding MARBERTv2 to the three-member word-only blend under tuned weights, to +0.24 with a 95% interval of [+0.03, +0.45].

The second track as an instrument. Two byte-identical files submitted to Open and to Strict returned identical metric vectors, so both tracks score the same 8,077-row blind gold. This makes the Open board a free measurement instrument for a Strict system whenever the Open runner-up is far enough behind that a losing probe costs no place. Two mechanics constrain the protocol, and both cost us before we understood them. First, we believed Codabench displays each participant’s latest submission; the competition API showed a submission rule of Add, so the board shows the entry the participant explicitly added and probing is close to free. Second, the board orders equal QWK by exact accuracy, so only a strictly better score should be sent to a track we lead. We record both because a system description that omits them would misrepresent how the final submission was chosen.

5 Results

5.1 Offline estimates against the board

Table 1 pairs every offline estimate we acted on with the live board reading of the same change. Two of the three reversed in sign, and the only change that delivered was the one with the smallest offline estimate.

The precision of an offline protocol is not its accuracy. Our nested out-of-fold measurements carry a standard deviation of 0.019 across grid settings, which invites reading the +0.406 MARBERTv2 gain as twenty standard deviations of signal; the board returned -0.1 . That 0.019 is the reproducibility of the procedure, not the uncertainty of the quantity it estimates. Nested cross-validation controls threshold overfitting and says nothing about distribution shift between the tuning and target samples, which is what separated validation from blind.

5.2 Saturation

Once the ensemble reached fourteen members it absorbed everything. Against an equal-weight baseline of 85.281 on honest nested out-of-fold, twenty fitted-aggregation configurations scored between 84.74 and 85.12: least-squares and ridge-shrunk weights, direct-QWK coordinate ascent, an expanded 44-member pool, per-member isotonic calibration, and median, trimmed-mean and rank-mean aggregation (Appendix C). Every one is below equal weight. Validation cannot support fourteen fitted parameters, and the row-wise extremes that robust aggregators discard carry real ordinal signal.

Member-level gains were absorbed too: a word-to-D3Tok lexicon worth +0.91 QWK solo moved the blend by between -0.024 and $+0.001$, because the cleaned member correlates more with the pack (Appendix C).

5.3 The endgame

With the mean provably capped, we changed the selection criterion. A dual gate, requiring a candidate to beat the baseline both on honest nested out-of-fold and on a deployment-mirrored oracle over the released gold test split, called four staged candidates statistical ties; probed on the board they returned 85.2, 85.2, 85.2 and 85.3, four for four at the board's resolution. So we stopped ranking candidates by offline score and ranked them by disagreement with the incumbent. Five near-clones, differing on 2% to 12% of rows, drew ties as well. The board ran out of resolution before our pool ran out of candidates. The test leg never overturned validation: its one promoted candidate failed the validation leg, and the board agreed.

5.4 Final standing

We finished second on both tracks with 85.3 QWK and 37.7 exact accuracy. Open ranked 4 submissions and the winner scored 85.5; Strict ranked 11 and the winner scored 85.4. The winner's exact accuracy exceeded ours on both tracks, 39.5 against 37.7 in Open, a difference our pool could not reach by recombination, calibration or retraining.

5.5 Correction to the submitted system

The entry we first submitted had four of its fourteen seats fine-tuned on the training split plus the corpus's released gold test split. We disclosed this to the organisers and in this paper, having read it as permitted; after the phase closed they ruled otherwise and we withdrew it. Every number here is the same fourteen-seat ensemble with those four seats replaced by their training-split-only counterparts from the same grid, an entry we also submitted during the testing phase and which the board scored at 85.3. What the withdrawn variant bought is measured in Appendix C: 0.008 offline and 0.1 on the board, inside every noise estimate in Section 4.

6 Conclusion

Three conclusions, in ascending order of how uncomfortable they are. First, a nested out-of-fold estimate is not automatically an honest estimate; ours was correctly specified and still overpromised by 0.5 QWK. Second, the reported standard deviation of a repeated measurement is not the uncertainty of what it measures. Third, on this task the leaderboard was the only instrument with adequate resolution, and the second track let us use it without risk. That last point is not general advice: it needs two tracks scoring the same gold and a margin wide enough to make a losing probe free. Teams without one should budget submissions as measurements rather than as attempts.

The counterfactual is worth stating plainly. Our own prioritisation document, written before these probes, ranked recovering the MARBERTv2 weights as the highest-value action remaining, on a +0.41 offline estimate. Following it would have shipped 84.7. If a seven-thousand-row holdout cannot resolve 0.2 QWK, some fraction of the ablation gains published for QWK-scored tasks, possibly including some in this paper, are reports of noise. Teams with board access should publish paired offline-versus-board comparisons.

Limitations

Our noise-floor evidence rests on three paired comparisons, enough to show that the disagreement occurs but not how often. Board readings are single measurements on a fixed 8,077-row set and carry their own sampling noise, which we could not estimate because we cannot resample the blind gold. The second-track oracle depends on a margin over the runner-up and was unavailable to us once that margin closed. Our locally computed D3Tok is systematically degraded relative to a licensed team's, and the standardisation fix recovers the member's usefulness without recovering that loss. Finally, equal weights and a single calibration scalar are unrefuted at our measurement resolution rather than shown optimal; we cannot rule out that a better combiner exists below our noise floor.

Ethical Considerations

BAREC frames sentence-level readability along a pedagogical reading progression (§2), so a system like ours is most plausibly consumed downstream to grade or select reading material for learners. We have only evaluated our ensemble on the shared task's own train, validation, test and blind splits; we do not know how it performs on registers, dialectal admixture, or genres outside that distribution, and a deployment decision (including or excluding a text for a given reader) should not extend our reported numbers to a population the corpus does not represent. The BAREC corpus and annotation guidelines we build on are cited in §2 and used under their released research licences; the D3Tok licensing asymmetry discussed above under Limitations is a data-access fairness concern between shared-task participants, not a claim about harm to end users of a deployed system.

Acknowledgements

We thank the shared task organisers. Fine-tuning ran on the National University of Singapore School of Computing compute cluster.

References

Ahmed Abdelali, Sabit Hassan, Hamdy Mubarak, Kareem Darwish, and Younes Samih. 2021. [Pre-training BERT on Arabic tweets: Practical considerations](#). *arXiv preprint arXiv:2102.10684*.

Ahmed Abdou. 2025. [mucAI at BAREC shared task 2025: Towards uncertainty aware Arabic readability assessment](#). In *Proceedings of The Third Arabic Natural Language Processing Conference: Shared Tasks*, pages 312–319, Suzhou, China. Association for Computational Linguistics.

Muhammad Abdul-Mageed, AbdelRahim Elmadany, and El Moatez Billah Nagoudi. 2021. [ARBERT & MARBERT: Deep bidirectional transformers for Arabic](#). In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 7088–7105, Online. Association for Computational Linguistics.

Muhammed Al Khalil, Nizar Habash, and Zhengyang Jiang. 2020. [A large-scale leveled readability lexicon for Standard Arabic](#). In *Proceedings of the Twelfth Language Resources and Evaluation Conference*, pages 3053–3062, Marseille, France. European Language Resources Association.

Bashar Alhafni, Reem Hazim, Juan David Pineros Liberato, Muhamed Al Khalil, and Nizar Habash. 2024. The SAMER Arabic text simplification corpus. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, pages 16079–16093, Torino, Italia. ELRA and ICCL.

Wissam Antoun, Fady Baly, and Hazem Hajj. 2020. [AraBERT: Transformer-based model for Arabic language understanding](#). In *Proceedings of the 4th Workshop on Open-Source Arabic Corpora and Processing Tools, with a Shared Task on Offensive Language Detection*, pages 9–15, Marseille, France. European Language Resources Association.

Wissam Antoun, Fady Baly, and Hazem Hajj. 2021. [AraELECTRA: Pre-training text discriminators for Arabic language understanding](#). In *Proceedings of the Sixth Arabic Natural Language Processing Workshop*, pages 191–195, Kyiv, Ukraine (Virtual). Association for Computational Linguistics.

Mohamed Basem, Mohamed Younes, Seif Ahmed, and Abdelrahman Moustafa. 2025. [!MSA at BAREC shared task 2025: Ensembling Arabic transformers for readability assessment](#). In *Proceedings of The Third Arabic Natural Language Processing Conference: Shared Tasks*, pages 297–305, Suzhou, China. Association for Computational Linguistics.

Alexis Conneau, Kartikay Khandelwal, Naman Goyal, Vishrav Chaudhary, Guillaume Wenzek, Francisco Guzmán, Edouard Grave, Myle Ott, Luke Zettlemoyer, and Veselin Stoyanov. 2020. [Unsupervised cross-lingual representation learning at scale](#). In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 8440–8451, Online. Association for Computational Linguistics.

Khalid N. Elmadani, Bashar Alhafni, Hanada Taha, and Nizar Habash. 2025a. [BAREC shared task 2025 on Arabic readability assessment](#). In *Proceedings of The Third Arabic Natural Language Processing Conference: Shared Tasks*, pages 239–252, Suzhou, China. Association for Computational Linguistics.

Khalid N. Elmadani, Bashar Alhafni, Hanada Taha, and Nizar Habash. 2026. BAREC-ST-2026: The second shared task on Arabic readability assessment. In *Proceedings of the Fourth Arabic Natural Language Processing Conference*, Budapest, Hungary. Association for Computational Linguistics.

Khalid N. Elmadani, Nizar Habash, and Hanada Taha-Thomure. 2025b. [A large and balanced corpus for fine-grained Arabic readability assessment](#). In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 16376–16400, Vienna, Austria. Association for Computational Linguistics.

Nizar Habash, Hanada Taha-Thomure, Khalid N. Elmadani, Zeina Zeino, and Abdallah Abushmaes. 2025. [Guidelines for fine-grained sentence-level Arabic readability annotation](#). In *Proceedings of the 19th Linguistic Annotation Workshop (LAW-XIX-2025)*, pages 359–376, Vienna, Austria. Association for Computational Linguistics.

Pengcheng He, Jianfeng Gao, and Weizhu Chen. 2023. [DeBERTaV3: Improving DeBERTa using ELECTRA-style pre-training with gradient-disentangled embedding sharing](#). In *The Eleventh International Conference on Learning Representations (ICLR)*.

Go Inoue, Bashar Alhafni, Nurpeiis Baimukan, Houda Bouamor, and Nizar Habash. 2021. [The interplay of variant, size, and task type in Arabic pre-trained language models](#). In *Proceedings of the Sixth Arabic Natural Language Processing Workshop*, pages 92–104, Kyiv, Ukraine (Virtual). Association for Computational Linguistics.

Fatimah Mohamed Emad Elden. 2025. [MorphoArabia at BAREC shared task 2025: A hybrid architecture with morphological analysis for Arabic readability assessment](#). In *Proceedings of The Third Arabic Natural Language Processing Conference: Shared Tasks*, pages 286–296, Suzhou, China. Association for Computational Linguistics.

Ahmad M. Nazzal. 2025. [ZAI at BAREC shared task 2025: AraBERT CORAL for fine grained Arabic readability](#). In *Proceedings of The Third Arabic Natural Language Processing Conference: Shared Tasks*, pages 266–268, Suzhou, China. Association for Computational Linguistics.

Ossama Obeid, Nasser Zalmout, Salam Khalifa, Dima Taji, Mai Oudah, Bashar Alhafni, Go Inoue, Fadhl Eryani, Alexander Erdmann, and Nizar Habash. 2020. [CAMEL tools: An open source python toolkit](#)

[for Arabic natural language processing](#). In *Proceedings of the Twelfth Language Resources and Evaluation Conference*, pages 7022–7032, Marseille, France. European Language Resources Association.

Mostafa Saeed, Rana Waly, and Abdelaziz Ashraf Hussein. 2025. [AMAR at BAREC shared task 2025: Arabic meta-learner for assessing readability](#). In *Proceedings of The Third Arabic Natural Language Processing Conference: Shared Tasks*, pages 320–330, Suzhou, China. Association for Computational Linguistics.

Xintong Shi, Wenzhi Cao, and Sebastian Raschka. 2023. [Deep neural networks for rank-consistent ordinal regression based on conditional probabilities](#). *Pattern Analysis and Applications*, 26(3):941–955.

Saoussan Trigui. 2025. [STBW at BAREC shared task 2025: AraBERT-v2 with MSE-SoftQWK loss for sentence-level Arabic readability](#). In *Proceedings of The Third Arabic Natural Language Processing Conference: Shared Tasks*, pages 362–366, Suzhou, China. Association for Computational Linguistics.

A Submission ledger

Every row is a live board reading on the blind gold. Rows marked with a dagger were probes uploaded to the Open track and read but never added to a leaderboard, which the competition’s Add submission rule makes free (Section 4).

system	calibration	QWK
word-only 3-member	$f = 0$	84.1
half-match champion	$f = 0.50$	84.5
match-60	$f = 0.60$	84.4
full distribution match	$f = 1.00$	84.0
metadata-reweighted prior	$f = 0.50$	83.9
blend4reg (both tracks)	$f = 0.50$	84.8
blend5, adds MARBERTv2 [†]	$f = 0.50$	84.7
blend4reg + doc shrinkage [†]	$f = 0.50$	84.6
fleet11	$f = 0.50$	85.1
fleet10strict	$f = 0.50$	85.2
fleet14 , shipped, both tracks	$f = 0.50$	85.3
fleet14 tie probes, 4 files [†]	$f = 0.50$	85.2 to 85.3
fleet14tt4, withdrawn	$f = 0.50$	85.4

Table 2: Blind-phase submissions, in the order they were measured.

Two rows in our internal log recorded a track that we could not re-confirm after the phase closed, and one calibration reading at $f = 0.40$ came from an ambiguous screenshot. The authenticated submission history is not reachable through the public API, so rather than guess we dropped the $f = 0.40$ point and describe the calibration peak as flat between 0.5 and 0.6 rather than across 0.4 to 0.6.

B Negative results in full

Metadata-reweighted label prior, refuted at 83.9. The half-match calibration matches the blind scores to the raw training label prior. That target is arguably wrong, because the blind set’s composition differs sharply from train: the STEM domain rises from 6.9% to 17.8% and the Specialized text class from 26.9% to 44.7%, while Advanced falls from 33.0% to 19.3%. Domain and text class are present for 100% of blind rows and take values that exist in train, which permits a label-free reweighting $P(\ell) = \sum_s P(s \mid \text{blind}) P(\ell \mid s, \text{train})$ over $s = \text{Domain} \times \text{Text_Class}$. This shifts the prior by L_1 distance 0.1224 and alters 1,697 of 8,077 predictions, every one by exactly +1 level. We built it as a strictly single-variable change: the implementation’s own raw-prior reference output is byte-identical to the previously submitted champion, so the prior is provably the only thing that differs. It scored 83.9, losing 0.6.

The hypothesis could not be tested offline, because blind has no labels and the test split does not exhibit the shift that motivated the correction. What the refutation rules out is specific: pushing 21% of predictions up one level makes things worse, so the blind set is not systematically harder in readability level than train despite being 2.6 times more STEM. We suspect this is because BAREC levels are annotated against a reading progression rather than topic difficulty, so domain composition and level composition are less coupled than intuition suggests, but we did not test that explanation.

MARBERTv2 as a member, measured at −0.1. MARBERTv2 was the most attractive remaining candidate on paper: a different backbone family, and among the least correlated with our blend at 0.945 against 0.95 to 0.98 for everything else. Offline it measured +0.41 on the clean test holdout and +0.406 on honest nested out-of-fold, the largest gain in either table. The test-holdout figure did not even reproduce offline: rerun from the cached arrays the baseline came out at 85.28 and the blend with the member at 85.28, a gain of exactly zero. Recovering it was expensive, because the original fine-tuning script saved only prediction arrays and never called `save_pretrained`, so the model existed only as numbers and had to be retrained. On the board it cost −0.1 in the five-member configuration, the only configuration con-

taining it that we submitted.

Document-level score shrinkage, measured at −0.2. The blind set ships a document column with 212 documents over 8,077 sentences, which invites smoothing each sentence score toward its document mean, $s \leftarrow (1-\alpha)s + \alpha \text{mean}_{\text{doc}}(s)$. On the clean test holdout the effect is single-peaked and sharp: 85.48 at $\alpha = 0$, 85.69 at 0.1, 85.32 at 0.2, 84.88 at 0.3, and a collapse to 82.15 at 0.5. We shipped the peak and it cost 0.2. One implementation subtlety has to be right for the transform to be the one that was measured: shrinkage must be applied to both validation and blind scores, because the rounder’s cutpoints live in validation score space.

Weak or redundant members. AraELECTRA regression is the most decorrelated candidate we measured at 0.9101, but at 78.29 solo it is too weak to help and moves the blend from 85.07 to 84.87. A D3Tok cross-entropy member costs −0.055 on honest nested out-of-fold and was excluded even though its sibling regression member is our best single addition. Decorrelation alone does not qualify a member; our working rule is that a member must be either near 85 solo or below 0.90 correlated, and one that is neither is not worth a seat.

C The saturation sweep

All figures are honest validation nested out-of-fold QWK, mean over threshold grids of 200, 300 and 400 candidates. The equal-weight fourteen-member baseline is 85.281 ± 0.051 .

aggregation	QWK
equal weight (shipped)	85.281
NNLS weights, ridge-shrunk 25/50/75	84.79 to 85.10
OLS and clipped weights	84.79 to 85.10
direct-QWK coordinate ascent	84.97, 84.98
expanded 44-member pool, NNLS	84.94 to 85.10
per-member isotonic calibration	84.74 to 85.12
median / trim1 / trim2 / rank-mean	84.79 to 85.01

Table 3: Twenty aggregation configurations, all below plain equal weight. Per-fold fitted weights were unstable, zeroing members in some folds and not others.

Member-level improvements were absorbed as well. We built a word-to-D3Tok lexicon from the gold corpus alignment (98.8% of sentences align), covering 92% of validation tokens and 91% of blind tokens, with the older disambiguation pipeline filling the remainder. This raised

sentence-exact fidelity against the gold column from 63.7% to 79.1% and lifted the affected regression member from 82.27 to 83.18 solo on validation, recovering most of the gap to gold text. At blend level the same change was worth -0.019 when swapped for the single member, $+0.001$ for the seed-averaged family, and -0.024 for both together. Retraining the five strongest families on the training split plus the released gold test split, 62,131 rows and a 13% increase in gold data, moved the blend from 85.281 to 85.289. That variant was withdrawn as outside the rules (Section 5.5). It is reported here because it is the cleanest evidence of saturation we have: 13% more gold labels moved the blend by 0.008 offline, and on the board the withdrawn file scored 85.4 against the shipped 85.3, on predictions differing for 811 of 8,077 sentences.

D Fine-tuning grid

Both tracks we entered are satisfied by the BAREC training split alone. We used no external readability resources: the SAMER corpus (Alhafni et al., 2024) and the SAMER readability lexicon (Al Khalil et al., 2020), listed for the Constrained and Open tracks, were not used, and we added nothing outside BAREC in the Open track either, since the 2025 evidence is that external data does not help on this task.

Training used a Slurm job array on A100 GPUs, with the single exception of the MARBERTv2 regression member discussed in Appendix B, which was fine-tuned in a free Kaggle notebook on a T4 and is not part of the cluster grid. One trainer script takes backbone, loss, input view, seed, learning rate and warmup ratio, and writes validation and test score arrays through the same code path used at inference. Defaults: learning rate 2×10^{-5} , 5 epochs, batch 16, maximum sequence length 128, no warmup except 0.06 for the large backbones. Cross-entropy, soft-label and focal heads are decoded to a continuous score by the posterior mean $\text{softmax}(z) \cdot [1..19]$; regression heads use the raw output; CORN follows Shi et al. (2023).

The dispatched grid has 112 rows over 13 backbones and 6 loss families (cross-entropy 45, regression 44, soft-QWK 10, focal 6, CORN 5, soft-label 2), crossing three input views (raw, Word, D3Tok) with seeds. All 112 completed with a done marker and both score arrays; a smoke-test run brings the total to 113. With public

checkpoints and seed-averaged families this produced a pool of 172 candidate members, 26 of them seed averages. The thirteen backbones are AraBERT (Antoun et al., 2020), AraELECTRA (Antoun et al., 2021), CAMELBERT (Inoue et al., 2021), QARiB (Abdelali et al., 2021), ARBERTv2 and MARBERTv2 (Abdul-Mageed et al., 2021), XLM-R (Conneau et al., 2020) and mDeBERTaV3 (He et al., 2023).

Two gates apply before a member may enter a blend, both added after an incident in which a checkpoint export was omitted and only prediction arrays survived. The head shape must match the fine-tune rather than the base model, and locally recomputed validation scores must reproduce the cluster’s saved arrays above 0.99 correlation. A member failing either gate is not blended, because a standardisation computed from mismatched scores is silently wrong rather than loudly wrong.

The fourteen seats of the final system are: the four public-checkpoint champion members (arabertv02-word-CE, camelbert-word-CE, qarib-barec-optimized-v8, arabertv2-d3tok-reg scored through our own D3Tok pipeline); seed-averaged families of arabertv2-reg-d3tok, arabertv02-ce-word, alarge02-reg-word and marbert-ce-raw; seed-averaged families of camelbert-ce-word, qarib-ce-raw, araelectra-reg-word, camelbert-reg-word and arabertv2-ce-d3tok; and a single xlm-roberta-large regression run. The four public checkpoints are CAMEL-Lab/readability-arabertv02-word-CE, CAMEL-Lab/readability-camelbert-word-CE and CAMEL-Lab/readability-arabertv2-d3tok-reg, released with the BAREC corpus (Elmadani et al., 2025b), and the community checkpoint AymanTarig/qarib-barec-optimized-v8, which has no associated publication and documents no training data.² That checkpoint entered our candidate pool through a screening script that scored each public candidate on validation and on the released gold test split, discarded anything above 0.88 on either as memorised, and retained only those above 0.80 on test. Its presence in the pool was therefore conditioned on a test reading. Nothing

²<https://huggingface.co/AymanTarig/qarib-barec-optimized-v8>

origin	heads / inputs	seats
public checkpoints	CE, reg; Word, D3Tok, raw	4
own AraBERT v02/v2/large	CE, reg; Word, D3Tok	4
own CAMELBERT, QARiB	CE, reg; Word, raw	3
own MARBERT, AraELECTRA, XLM-R	CE, reg; raw, Word	3

Table 4: The fourteen seats, by origin. Nine are seed-averaged families of our own fine-tunes, four are public checkpoints and one is a single run. All are trained on the BAREC training split. The exact member list is given below.

downstream is fitted on test, and a validation-only greedy over the compliant pool selects it first of 67 candidates, so the gate was not what put it in the system.

Every seat we fine-tuned ourselves uses the BAREC training split alone. The four that did not, in the entry we first submitted, were replaced by their training-split-only counterparts from the same grid after the organisers ruled on them (Section 5.5). The nine seed-averaged families and the single run each carry a per-run training configuration on disk, and none names any split other than train. The four public checkpoints were fine-tuned on BAREC by others: three are the corpus authors’ own baselines, whose paper describes fitting on train and selecting on dev, and the fourth is a community checkpoint whose author documents no training data at all, which we report because we cannot verify it.

E Harness detail

The rounder maps a continuous score to a level in $[1, 19]$ via 18 monotone cutpoints chosen by coordinate ascent to maximise QWK. The search is seeded from equal-frequency quantiles of the label distribution, each cutpoint is swept over a fixed grid spanning the score range, monotonicity is enforced between neighbours, and the output is clipped. We use monotone coordinate ascent rather than Nelder-Mead, which can return crossed, non-monotone thresholds in 18 dimensions and is retained only as a cross-check. On a classical ridge baseline the protocol reads cleanly: naive rounding 71.25, nested out-of-fold 74.48, non-nested out-of-fold 74.63.

Folds are built once with grouped stratified 5-fold cross-validation, grouped on document and stratified on the 19-level label, and serialised to a committed document-to-fold map covering 1,518 documents. The map is loaded identically on every machine and never regenerated, because regenerating it per machine is how out-of-fold arrays

silently fail to line up. All five folds carry all 19 levels.

The parity gate compares our scorer against the official development fixture at the level of the printed string: QWK 80.0040%, accuracy 56.6211%, adjacent accuracy 69.8632%, average absolute distance 1.143776, and 65.8687% / 70.2736% / 76.4569% at 7, 5 and 3 collapsed levels.

F Reproducibility

Code, the fine-tuning grid, and the per-run validation and test score arrays are at <https://github.com/thylinao1/barec-2026-readability> under an MIT license. Rebuilding the final submission from the committed blender and the member score arrays reproduces the submitted file byte for byte; its SHA-256 is `cf2c752406a9ea2bc0ad94e83f40190cab87b958512ef9c03dcf01c12768b762`. The member arrays computed over the blind input are withheld together with everything else derived from the confidential blind set, so the rebuild command in the repository README needs either those arrays or the blind input to regenerate them.

Predictions over the blind test set are deliberately not published. The blind set is released only to registered participants and may be reused in a future edition, so a strong prediction file over it would function as a pseudo-label source for later participants. The code regenerates those files for anyone holding the blind input, and the hash above lets the organisers verify the submitted file exactly.